

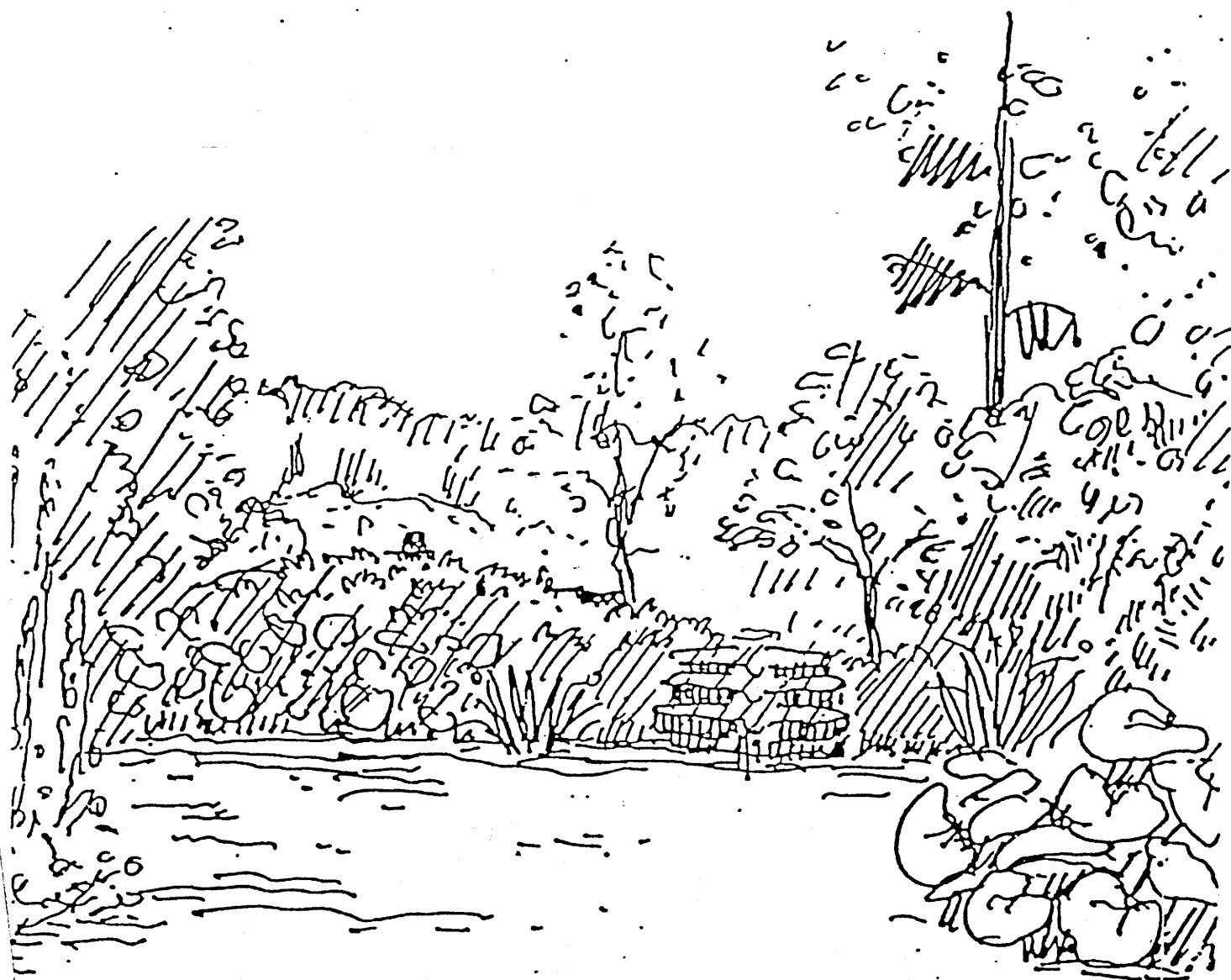
THE BIOLOGICAL CLEANSING DAMS

AT THE

RUDOLF STEINER SEMINARIET

JÄRNA

USING VIRBELA FLOWFORMS



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Translated from the Swedish by:

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One of the fundamental ideas with water treatment work in Jarna is that we as people must take consideration to the larger cycles in nature

eg: The Water Cycle

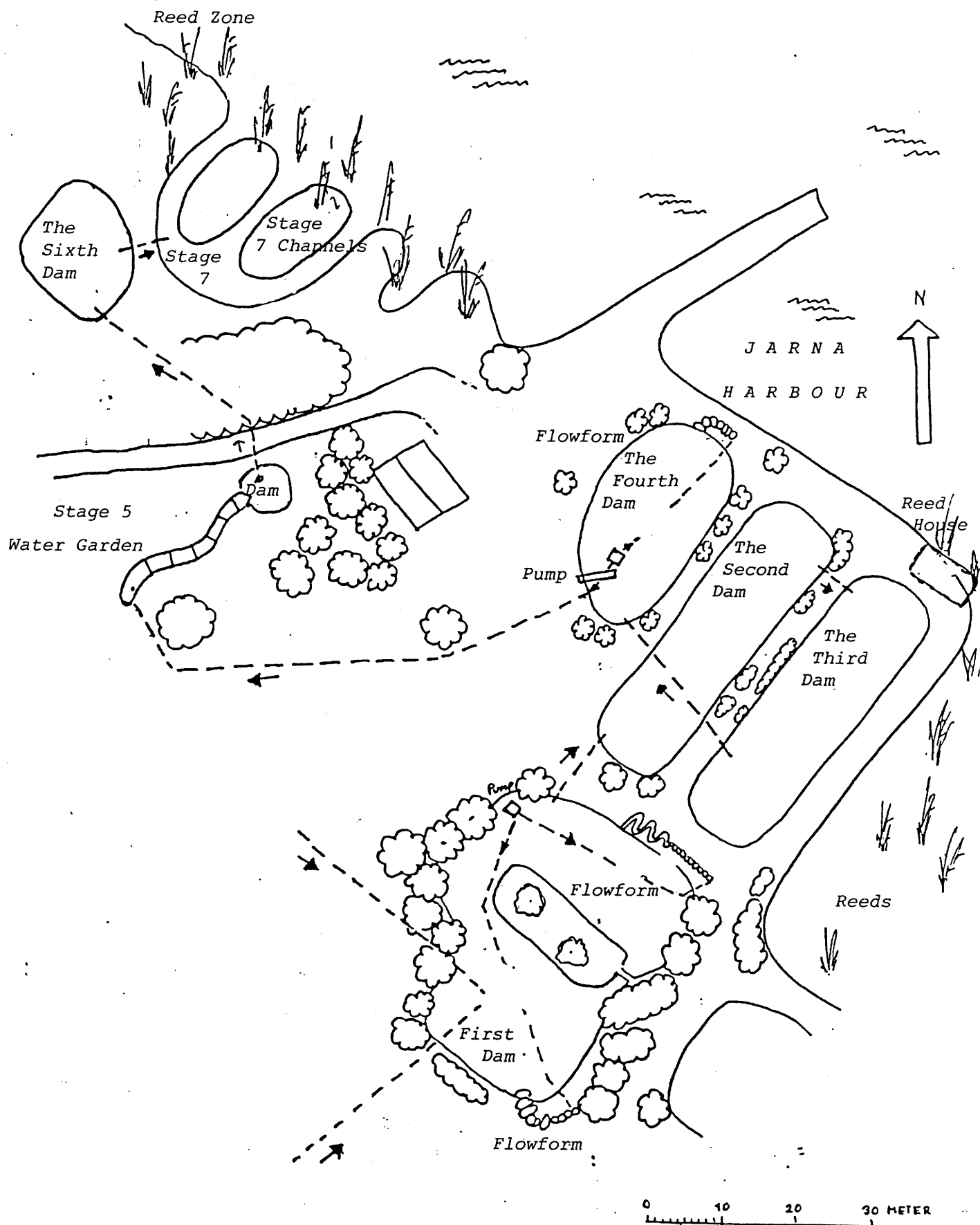
Therefore, Sewage does not disappear just because we eg: flush it down the toilet.

The task of transforming the pollutions from a burden to a resource.

The dam system and its surroundings form a pleasant park (the sewage fertilises the vegetation) and are an often used recreational area.

Flowforms set up by the First Dam to recirculate effluent water.





The Rudolf Steiner Seminaries

Waste Water/Sewage Treatment Plant

Description

The Rudolf Steiner College Wastewater treatment plant is situated in the belt of reeds in Jarna Bay (an inlet of the Baltic) as part of the Seminaries gardens.

It has been incorporated in the gardens and is now part of the scenic walk.

The system receives sewage from 75-100 people.

Only easily degradable detergents are used in the houses. The dams were built in 1972.

The system consists of eight different steps: Five different dams, a water garden and a channel system through the reeds and out into the harbour.

It takes approximately one year for the sewage water to pass through the system.

In a dried out area (reed area) four new dams have been added to the first original dam in the garden.

The systems total area is about 2300 square metres. Therefore:- about 20 square metres per person.

Stage One

The first dam is a "breaking down" dam of about 900 square metres with an average depth of 1.5 metres.

Two Flowforms belong to this dam.

In a period of 5 days as much water as the dam holds passes through the flowforms.

It takes about 100 days for the water to flow through the first dam.

This dam receives totally untreated sewage water. The organic material is broken down by anaerobic and aerobic bacteria, which provide food for, among other things, "Rotatories". Plenty of cilia and flagellia also occur.

The "Cyano" bacterias photosynthesis provides an important addition of O₂ to the dams' upper layer.

A part of the (Hydrogen Sulphide) (H₂S) formed by the anaerobic "breaking down" is oxidised by "Thio" bacteria (transport sulphur bacteria) especially during the winter half of the year.

The colour of the dam changes, depending on the competitive relationship between the different organisms, from grey to brown, olive green and dark green.

The first dam is surrounded by flourishing greenery. Mostly large leaved plants such as "Bear Lily", Pestilen Root and Coltsfoot and plants which need high nourishment such as "Stinging nettles" etc.

These plants collect nourishment from the waters edge around the dam.

The organic material is stored at the bottom of the dam during the winter, and later "broken down" during the summer half of the year (frozen for 5 months of the year).

The Flowforms continue to flow even under the layers of snow and ice.

Surface ice on the dam can be up to 40cm thick.

Stage Two

Through surface runoff the water is carried over to the second dam which is characterised by plentiful algae growth.

The dam is 450 square metres x 1.70 metres deep. The algae present are chlamydomonus, (Chlorophycene) Euglena, Phacus, and Trachelomonas. All three belonging to the Euglenaceae family.

They grow and produce so much oxygen that small bubbles are formed. The algae float up to the surface on the small bubbles and form a thick algae layer over the whole dam.

The first algae "Flowering" usually happens during May and June followed by a period of summer stagnation.

A second algae "Flowering" occurs in August-September. The plants around the dams edge mainly consist of different types of reeds - see diagram (Vass, Kolvass). Sometimes the plant "Bittersweet". The "Kolvassen" and different algae are believed to have a bacteria "killing" effect hence the low numbers of "coli" bacteria in the H₂O when it has passed this dam.

Stage Three

The third dam is the same size as No 2 and is called the "Animal" dam because of the high number of "membrane crayfish" present.

It is surrounded by a reed belt (*Phragmites Australis*).

In this dam we try to achieve a wide variety of plants and hence have dug out some reed roots and planted other "Rushes" types and kaveldun (see illustration Page). We have also tried duck weed (*Lemna minor*) which has spread over the whole dam surface. The dam's ducks eat this but sometimes the growth is so vigorous that the dam must be weeded.

Stage Four

The fourth dam is smaller (300 square metres) with a sloping bottom 0-1.4 metres.

A Flowform has been joined to this dam. Carp have been introduced, we have yet to see how they like it.

H₂O in dam 4 is so clean that a rich plant and animal life has been able to develop.

The amount of Coli Bacteria is so low that the "State" could allow swimming.

One would like to increase the diversity of animal and plant life.

The reed belt has been partially replaced with other plants, "kaveldun" waterlily, sword lily and types of sedge etc.

Animals, "tummy swimmers", "back swimmers", divers, dragonflies, water snails, newts, "water bats", eels, and ducks.

Stage Five

Is a 25 metre long Water Garden with different water plants. eg: Malcadan and other before mentioned plants.

It has been built so that the water runs alternatively through and over stones, sometimes small shallows are formed.

Stage Six

Is a smaller dam (130 square metres) lying in the Reed belt next to the channel system.

Stage Seven

The seventh stage is a dug out channel system through the reed belt with a total length of 75 metres by 1.5 - 2.0 metres wide. One end is connected with the Jarna Bay.

The material which was dug out to form channels, has been used to form small islands where Elm, Silver birch and salix family plants have been planted as shelter for nesting birds.

Fry (Baby Fish) are found, (all the way through the channels), through placing reed bunches outside the belt for fish to lay eggs in, and through cutting the reeds within the channels that are not yet dried up.

One tries to make it possible for fish and bird life to inhabit the area.

The effect of the "sewage" water on the inlet are also studied here.

Stage Eight

The 8th stage is the Jarna Bay.

If one works with such a water treatment system and wishes to release the water out into the harbour one must also think of the sea as part of the system.

An active "Watercare" in the Jarna Bay is also part of the work with the dams.

Observation of the changes in Flora and Fauna must go hand in hand with measures to better conditions for fish and bird life in the Bay.

That is why one has made the abovenamed Rush bunches or Reed bunches for fish egg fertilisations and Fry (small fish) protection.

On the stakes used to hold the Reed bunches in place small platforms have been made which may act as nesting places for the "Grass Ducks".

The Flowforms

Two connected to the first dam and one Flowform connected to the fourth dam.

Before the incorporation of the Flowforms into the system it did not function satisfactorily, eg: "smells" rising mostly from above the first dam.

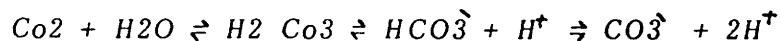
The water is taken from the bottom of the ponds to the flowforms hence creating a circulation.

As the H₂O runs through the Flowforms several things happen:

(1) Bad smelling gasses, eg: Methane, H₂S, Ammoniae etc that are formed by the anaerobic "breaking down" at the bottom of the pond, are aired out.

(2) Co₂ is driven out. This displaces the Carbonic Acid balance towards the left.

The result of this is a raised PH value.



(3) The water is oxygenated.

After the installation of the flowforms Cyano bacteria (blue-green algae) growth rate was increased greatly in the first dam.

Cyanobacteria oxygenates the waters overlayer making possible aerobic breakdown.

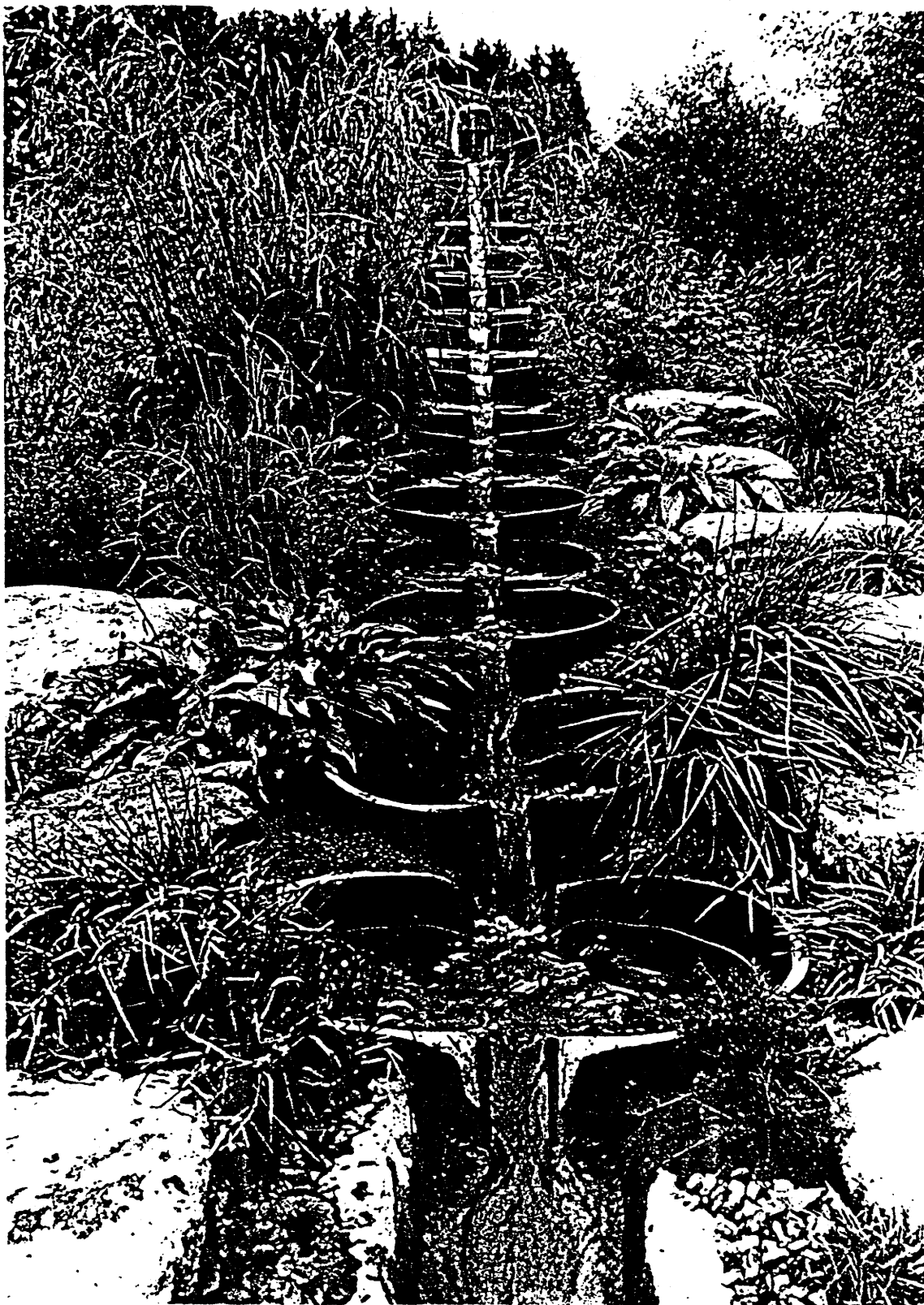
John Wilkes an English Sculptor constructed the first Flowforms in 1970.

Up until 1983 hundreds of Flowforms have been erected, mostly as artistic ornamentation in Parks etc.

Research into Flowforms is being carried out on a limited scale in European countries, Great Britain, West Germany, Holland, Switzerland among others as well as in U.S.A., Australia, New Zealand under the name Virbela Flow Design Research Group. (1991 - ~~now~~ ECOLOGICAL DESIGN and RESEARCH INSTITUTE)

Among other things researchers are trying to ascertain what effect Flowform "treated water" has on different living organisms.

Flowforms set up by the first pond to recirculate effluent water.



Plants as Water Cleansers

At the Stiftung Limnologische Arbeits Group in Krefeld-Hulzenberg, West Germany, Doctor Kathe Seidal has carried out research for the last 30 odd years, on how plants can be used in conjunction with water cleansing.

It has been discovered that the two most interesting plants are (1) "Kolvass" (*Scirpus lacustris*) and (2) Reeds (*Phragmites australis*).

Plant (1) occurs mostly within the vegetation around the second dam and plant (2) dominates in the channels and along the Jarna Bay shoreline.

It is important that both (1) and (2) effectively dry out and mineralise the Sludge. In addition pathogenic bacteria, *Salmonella*, *Enterokocker*, *Escherichia Coli* and Worm eggs are killed.

We are yet to discover how this happens. Plant (1) bacteria killing effect is important for the second dam's function.

Several other interesting observations have been made about Kolvasse (1). It effectively absorbs certain heavy metals, eg copper, chrome, nickel, vanadium and manganese, out of the sewage water.

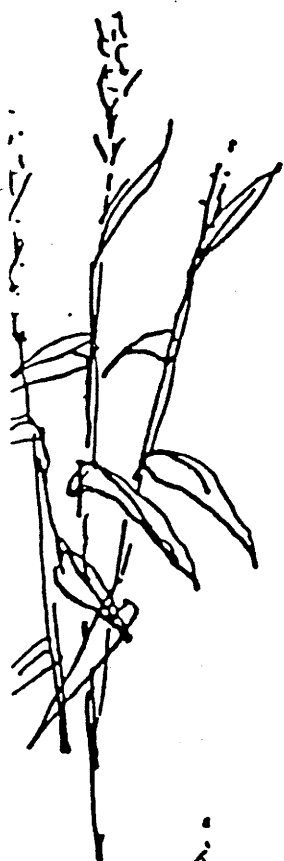
It shifts H₂O pH to neutral and converts certain poisonous substances, eg Fenols and cyanides into non-poisonous substances in the same way one's own body does.

The course of events of the chemical during the conversion of the Fenols has been worked out and is clear.

At a camping place near Edberg in Holland the sewage water from up to 6,000 campers is treated with a channel system, in which Reeds and Kolvass (1) have been planted. It takes the water 10 days to flow through the system. It is so clean after going through this system that it can be used as drinking water for stock.

The plants are harvested regularly and dried. They are then sold as a weaving material and the resulting income makes the system self-sufficient.

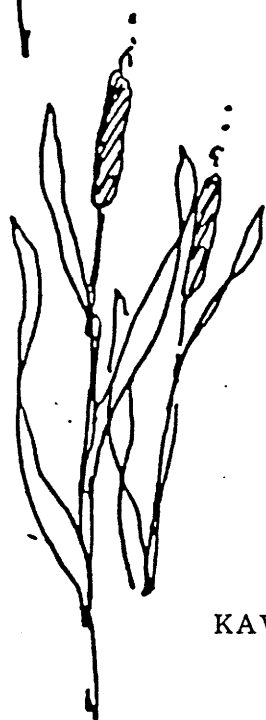
Plants as water cleansers are most suited in areas with long vegetation periods.



REEDS



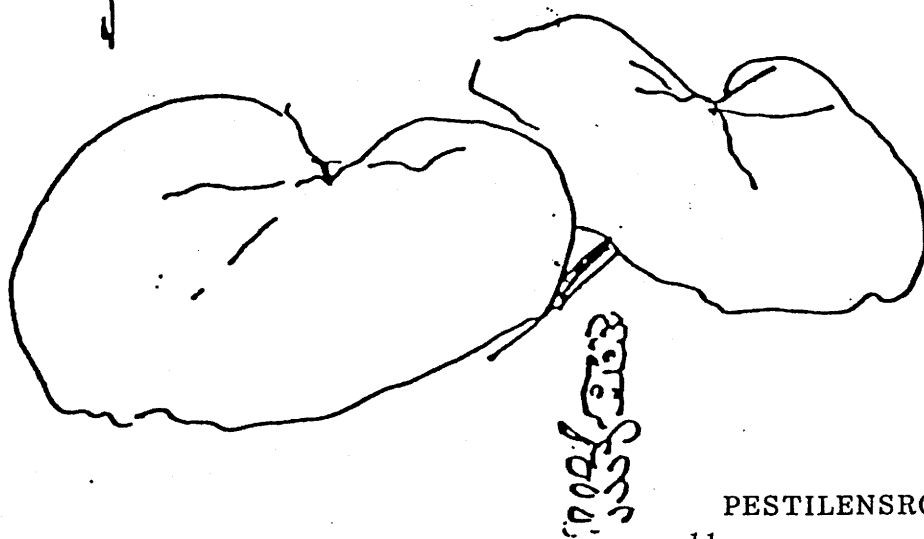
KOLVASS



KAVELDUN



SWORD LILY



PESTILENSROT

Measurements

The water that is let into the dams is "household" sewage/waste water.

ie: Comes from separate households, big kitchens, communal laundries, bathroom and toilets.

A lot of the scraps from food preparations are saved for compost.

A washing powder with a low level of Phosphate is used in the washing machines.

Sewage/waste water is not allowed to have any non degradable waste as no sorting/filtering is done before the water enters the first dam.

The load can vary greatly depending on whether it is a school term or not, etc.

When the system was new, tests were taken every second week. Now they are taken every fourth month.

Analysis is made of pH, BS7, Phosphur (Organic and non-Organic) Nitrogen, Electric conductivity. Measurements vary according to pressure on the system and time of year.

The following diagram shows the results from measurements of pH, BS7, Total-P and Total N under the whole or parts of February 1980 - May 1986.

The pH is never lower than 7 in spite of acid (sour) rain and "sour burps" which occur in conjunction with snow melting.

In the last dam the pH (average pH7.5) is a little higher than in the first dam (average pH 7.3).

The average value of BS7 sinks from 105g/m³ in the first - 26g/m³ in the last dam.

The average cleansing effect is therefore 75%.

The total phosphor content in the first dam has an average of 8.6g/m³. In the last dam it decreased to 4.7g/m³.

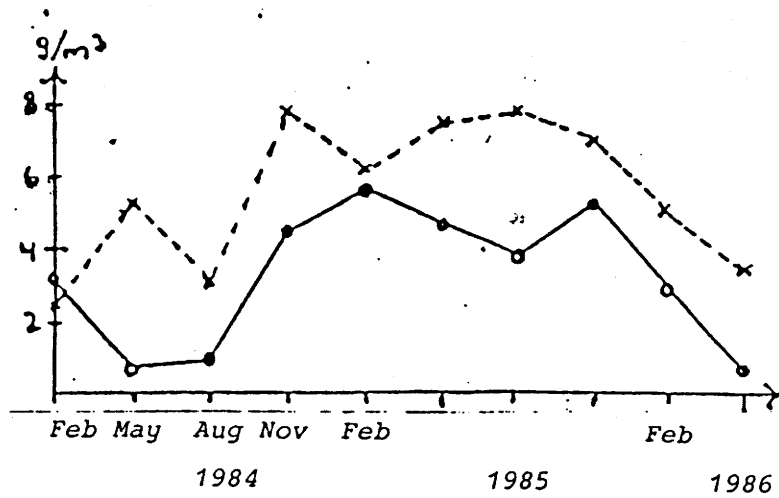
The average cleansing effect is 45%.

The total Nitrogen content has the comparative measurements of 5.5g/m³ in the first dam and 3.2g/m³ in the second dam. The average cleansing effect is 42%.

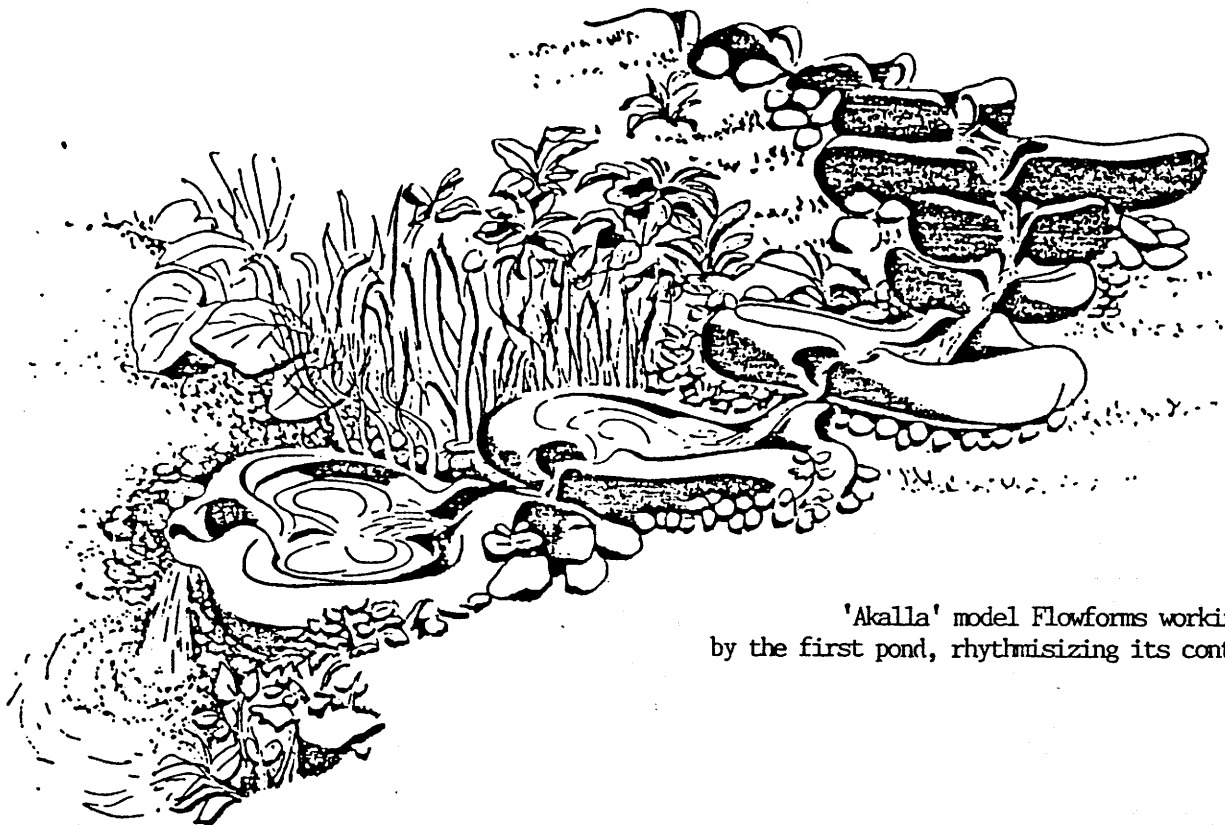
See chart page 13.

x = First Dam
o = Second Dam

Nitrogen Content



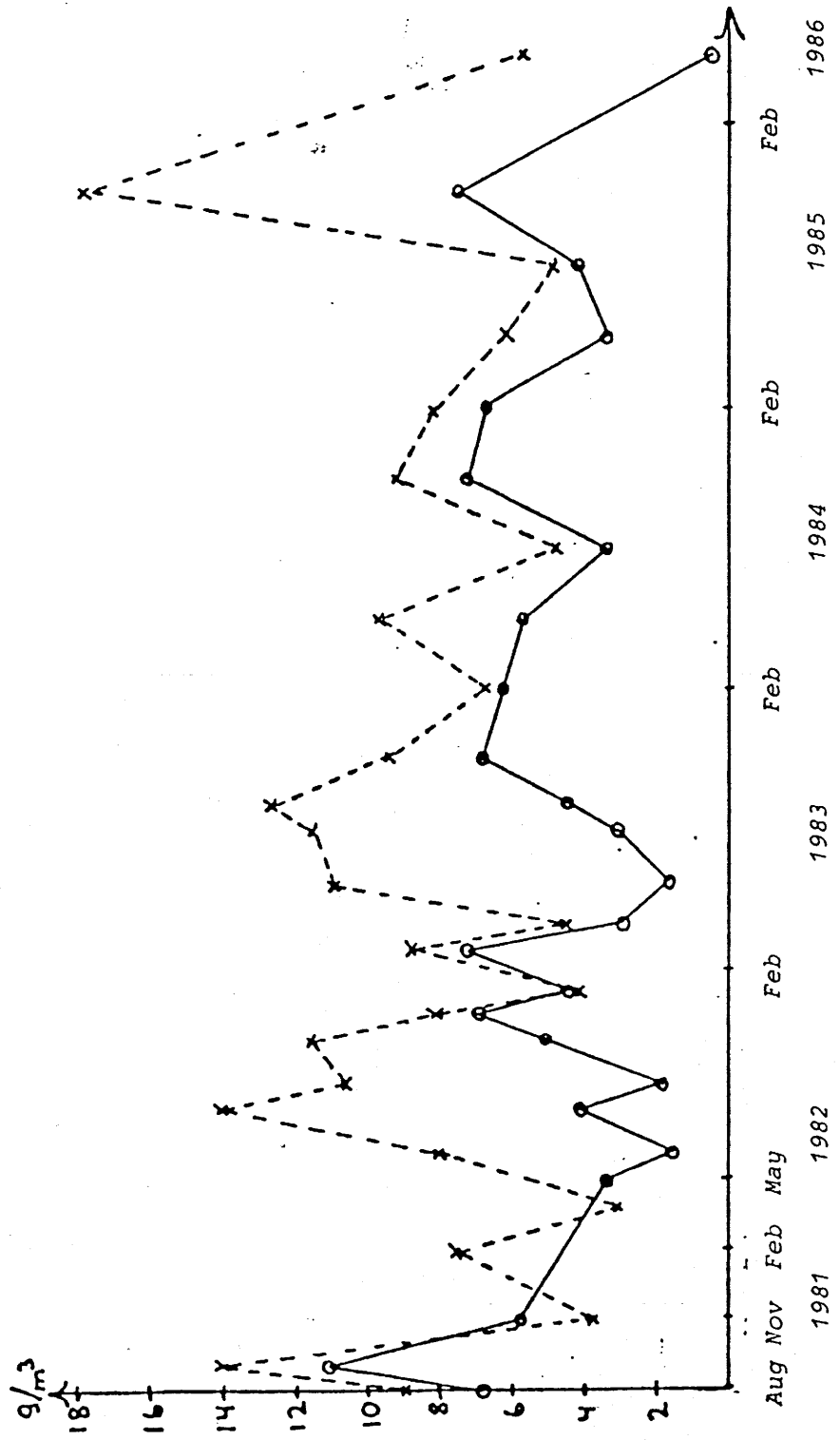
Average x = 5.5 SD = 2.0
o = 3.2 SD = 1.9



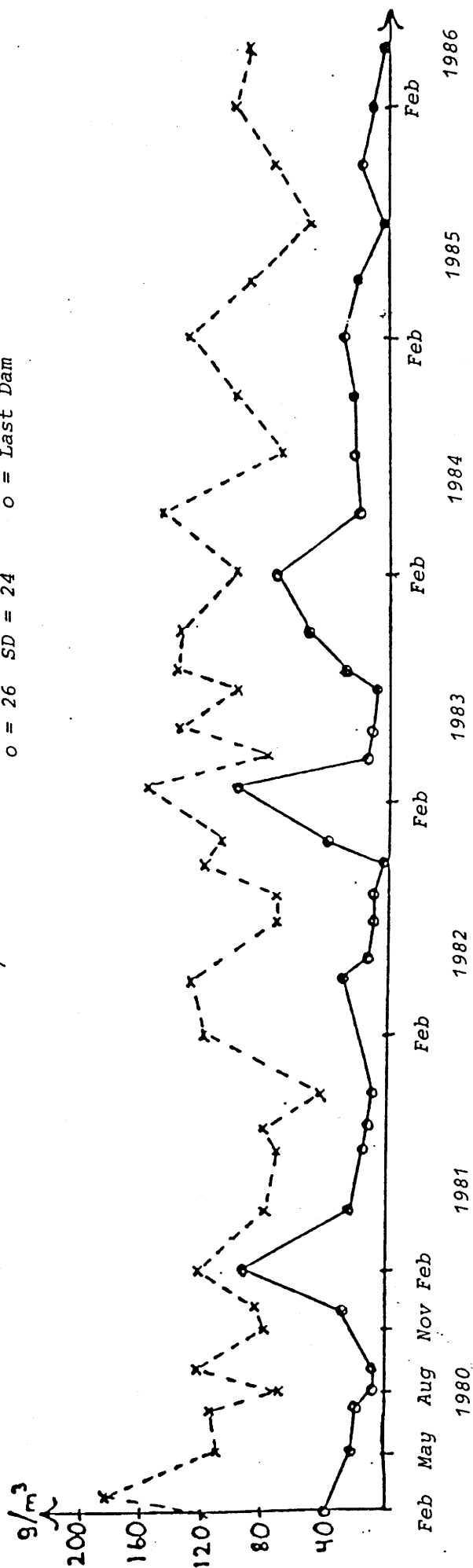
'Akalla' model Flowforms working
by the first pond, rhythmisizing its contents.

TOTAL PHOSPHORUS LEVELS

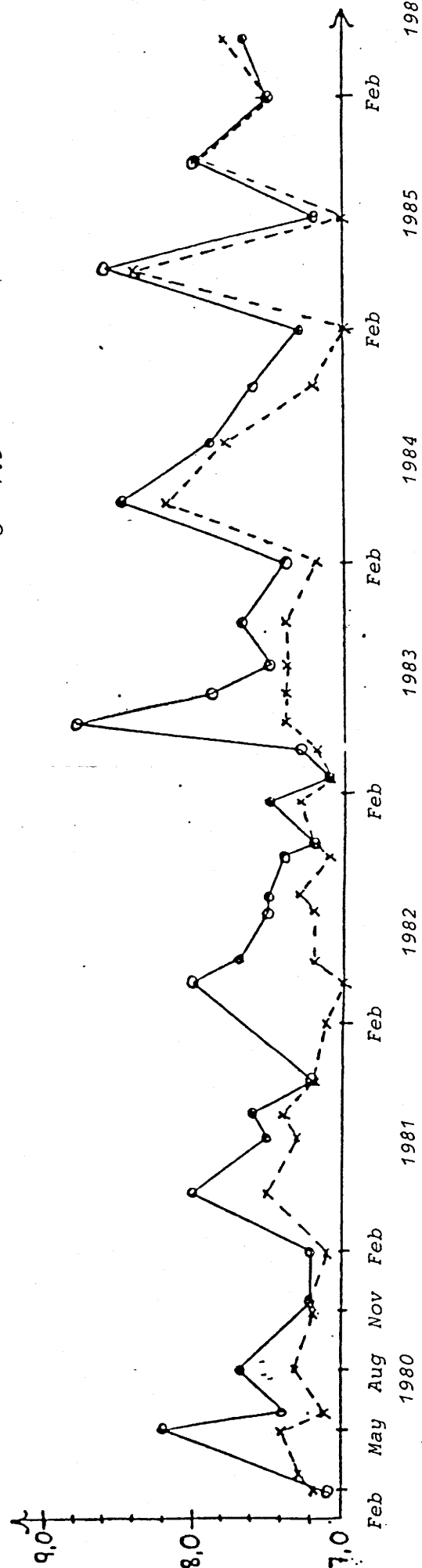
Average $\bar{x} = 8.6$ $SD = 3.6$
 $\bar{o} = 4.7$ $SD = 2.4$



BS 7



Average $x = 7.3$
0 = 7.5



Comparison With Other Cleansing

Such a sewage treatment system as described above can partially be compared with the cleansing in three stages (Mechanical, Biological, Chemical) method often used at larger treatment stations and partially compared with resorbtion or infiltration systems which are used mostly with smaller amounts if outlet is from only a few properties.

In large Treatment Plants (or works) at least 90% of the phosphor must be removed from the water.

Research is being done to develop effective methods to remove Nitrogen, these are not yet in use in treatment plants.

The interest in removing nitrogen has increased greatly since research has shown that Nitrogen probably plays a greater role in the eutrofication of the Baltic, than Phosphor.

BS7 is brought down after to a very low level. A problem with this type of treatment plant is the "Rot Sludge" which is left.

There one finds again the polluting substances in concentrated form.

Treatment plants have great difficulty in getting rid of the "Rot Sludge" but is often used as fertiliser hence returning nutritive salts, heavy metals etc into nature.

A system of biological dams is not any alternative in built-up areas (urban). The biological processes are disturbed or made impossible by oil and chemicals from for example, industry, and a dam system demands a large area and can only take a limited amount. In a reabsorbtion system untreated water is spread through the upper soil layers and taken up mostly by vegetation.

An infiltration plant consists of a 2 or 3 chamber "well", where the sludge sedimentates, and pipes/hoses through which the water is spread through holes.

p17 is missing

The water then filters down to the ground water. Sludge is emptied and "rotted" later, most often at treatment stations.

A dam system should be able to be used instead provided of course that a water way or harbour (river etc) can receive the water. If the dams are combined with a "decay/Rotting Room" silting up is decreased as the amount of larger particles in the sewage water is less.

The Rotted waste from a "Rot Room" (chamber) is furthermore easier to take care of than bottom sludge.

A Study Piece

The Seminaries treatment system has given inspiration to Painters, Sculptors, Biologists, Botanists, School pupils, Poets and Musicians, that, during the years, have visited and studied the dams.

Both Scientists and Artists can wander around there and experience a unity of Art and Science forming a human wholeness - something that is both beautiful and functional.

Lennart Olsson and Nigel Wells
Drawings - Ingvar Lonnstedt

July 1986

Literature

This presentation has been built mostly on a project carried out at Stockholm University by Arja Halvar, Kristina Lonn and Lennart Olsson.

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